CHM102 Basic Organic Chemistry

Spring semester

January 2025

Instructor: Dr Sugumar Venkataramani, Office: 5F10, AB2

Tutors: Dr Moitree Laskar, Dr Pooja & Dr Tamanna

Schedule

Lecture timing

Tuesdays 12.00 to 1.00 pm
Venue: LH6

Wednesdays 3.00 to 4.00 pm
Fridays 3.00 to 4.00 pm
Venue: LH5

I Mid-sem exams

February 6, 7 & 8, 2025
CHM102: Feb 8, 2025 – 10 am

II Mid-sem exams

March 6, 7 & 8, 2025
CHM102: Mar 8, 2025 – 10 am

End-sem exams

April 28 – May 10, 2025
CHM102: May 1, 2025 – 2.30 pm

Day/Month	January	February	March	April	May
Tuesday	7 Lecture-1	4 Lecture-12	4 Lecture-23	1 Lecture-31	
Wednesday	8 Lecture-2	5 Lecture-13 1st Mid Sem Exams **(only for premajors in the afternoon session)**	5 Lecture-24 2nd Mid Sem Exams **(only for premajors in the afternoon session)**	2 Lecture-32	1 End sem exams 2 End sem exams 3 End sem exam **4** 5 End sem exams
Thursday	9 Last date of late registration	6 1st Mid Sem Exams (all batches)	6 2nd Mid Sem Exams (all batches)	3	6 End sem exams
Friday	10 Lecture-3	7 1st Mid Sem Exams (all batches)	7 2nd Mid Sem Exams (all batches)	4 Lecture-33	7 End sem exam 8 End sem exams
Saturday	**11**	8 1st Mid Sem Exams (all batches)	**8** 2nd Mid Sem Exams (all batches)	**5** MS thesis open seminar/ poster session	9 End sem exams 10 End sem exams
Tuesday	14 **Makar Sankranti**	11 Lecture-14	11 Mid-term break	8 Lecture-34	Deadline to show corrected answer sheet for all batches excpet premajors
Wednesday	15 Lecture-4	12 Lecture-15	12 Mid-term break	9 Lecture-35	
Friday	17 Lecture-5	14 Lecture-16	14 **Holi** Mid-term break	11 Lecture-36	
Tuesday	21 Lecture-6	18 Lecture-17	18 Lecture-25	15 Lecture-37	
Wednesday	22 Lecture-7	19 Lecture-18	19 Lecture-26	16 Lecture-38	
Friday	24 Lecture-8	21 Lecture-19	21 Lecture-27	18 **Good Friday**	
Saturday	**25**	**22**	**22**	**19 Lab Exams (Friday time-table for all courses)**	
Tuesday	28 Lecture-9	25 Lecture-20	25 Lecture-28	22 Lecture-39 Lab Exams	
Wednesday	29 Lecture-10	26 Lecture-21	26 Lecture-29	23 Lecture-40 Lab Exams	
Friday	31 Lecture-11	28 Lecture-22	28 Lecture-30	25 Lecture-41 End sem Exams (for premajors) (MS Thesis (PRJ502 & Int.PhD MS thesis) final hard copy submission Deadline)	
Monday			31 **Idu'l Fitr**	28 End sem exam	
Tuesday				29 End sem exam	
Wednesday				30 End sem exam	

Total – 41 lectures

{Before I midsem: 13 lectures; Between I & II mid-sem: 11 lectures; After II mid-sem: 17 lectures}

Grading Policy

	Plan A	Plan B
I – Mid-sem exam	20	15
II – Mid-sem exam	20	15
End-sem exam	50	50
Assignments	10	5
Surprise quizzes/ Attendance	0	15

***Plan B will be activated if the attendance drops significantly**

CHM102: Basic organic chemistry [Cr:3, Lc:2, Tt:1, Lb:0]
Course Outline

- A basic introduction to the importance of organic molecules in life.
- Nomenclature of organic compounds with diverse structure and functional groups.
  Concepts involving acidity, basicity, tautomerism, inductive, mesomeric, Nathan field and steric effects.
- Stereochemistry: Conformational analysis of acyclic (ethane, butane, substituted acyclic compounds) and cyclic (cyclohexane, cyclopentane) systems. Energies and conformational preferences of differently substituted cyclohexanes and related systems.
  Axial (atropisomerism) and planar (helical) chirality.
  Optical isomerism in biphenyls, allenes.
  Geometrical isomerism (E/Z), acylic and cyclic compounds.
  Description of syn/anti (erythro, threo), meso isomers. Absolute and relative stereochemistry.
  R/S configuration assignment. Racemization, resolution, prostereoisomerism, stereotopicity and enantiomeric excess, d,l isomers and racemate.
- Reaction types and basic reaction mechanisms involving:
  Substitution reactions: Nucleophilic substitution at carbon and its mechanism (including chirality generation).
  Grignard types reagents and their addition to carbonyls.
  Addition reactions: Addition of X, HX, boranes and hydroxylation to C=C systems and epoxidation of olefins.
  Stereoaspects of the addition of X, HX, X_2, boranes and hydroxylation to C=C systems. Cis- and trans-hydroxylation of cycloalkenes and related reactions.
  Elimination reactions: Elimination reactions leading to C=C bond formation and their mechanisms including named reactions (e.g., Hoffman and Saytzeff eliminations).
- Selectivity in organic reactions: Chemoselectivity, regioselectivity, enantio- and stereo-selectivity (explanation with examples).
- Reactive intermediates: Carbenes, nitrenes, carbocations, carbanions, and radicals. Reactions involving reactive intermediates, rearrangement reactions and mechanisms.
- A basic introduction to thermodynamic vs kinetic control, Hammond postulate (energy profile diagram).
- Aromaticity and anti-aromaticity: Hückel and Craig's rules, homo, hetero (furan, thiophene and pyrrole) and non-benzenoid aromatic systems. Polyaromatic compounds.

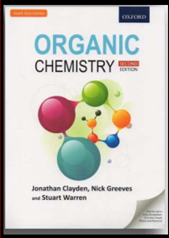
CHM102: Basic organic chemistry [Cr:3, Lc:2, Tt:1, Lb:0]

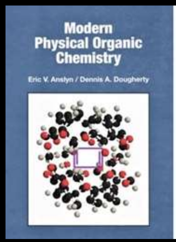
Recommended Reading

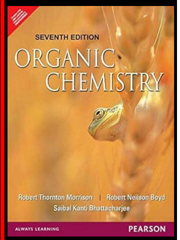
- J. Clayden, N. Greeves, S. G. Warren, *Organic Chemistry*, 2nd Ed, Oxford University Press, Oxford (2012).
- R. T. Morrison, R. N. Boyd, S. Bhattacherjee, *Organic Chemistry*, Indian Edition, 6th Ed, Pearson Education, New Delhi (2007).
- E. V. Anslyn, D. A. Dougherty, *Modern Physical Organic Chemistry*, University Science Books, 2005.

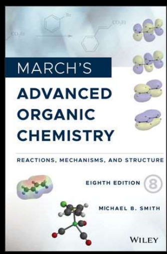
Suggested Reading

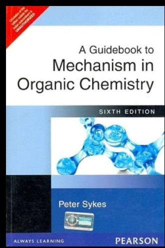
- M. B. Smith, *March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure*, 8th Ed, John Wiley & Sons, New Jersey (2020).
- P. Sykes, *A Guidebook to Mechanism in Organic Chemistry*, Indian Edition, 7th Ed, Pearson Education, New Delhi (2010) (Dorling Kindersley (India) Pvt Ltd).

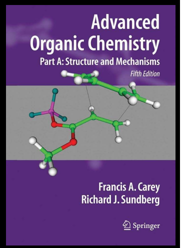






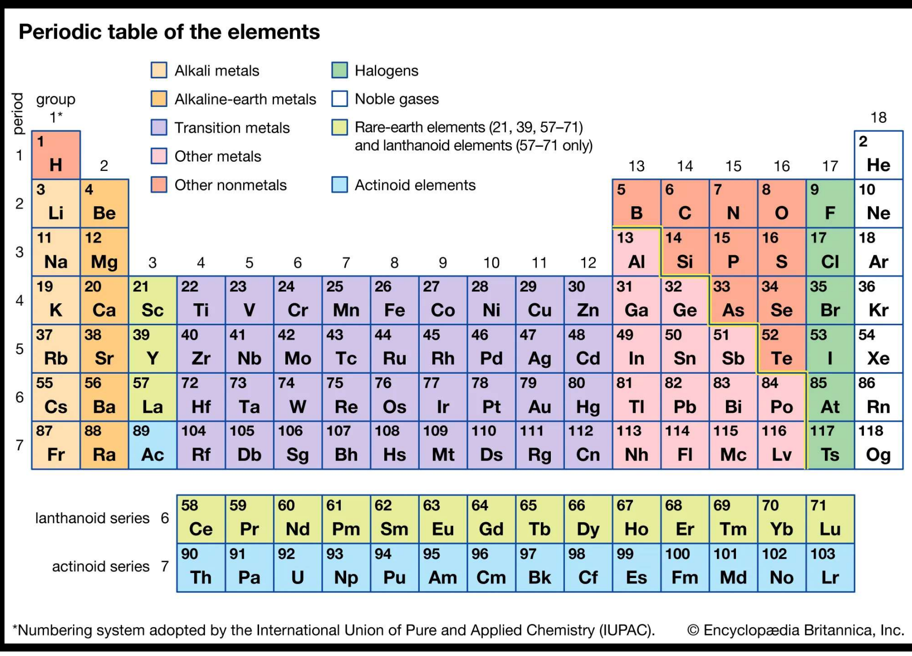




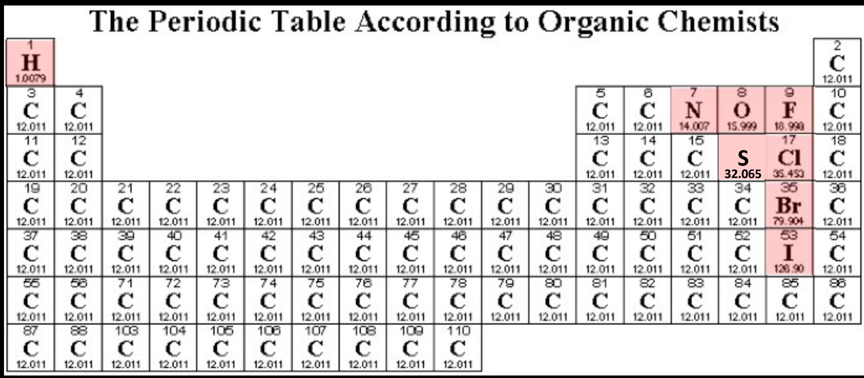


Part A and B

Why organic chemistry?



Why organic chemistry?

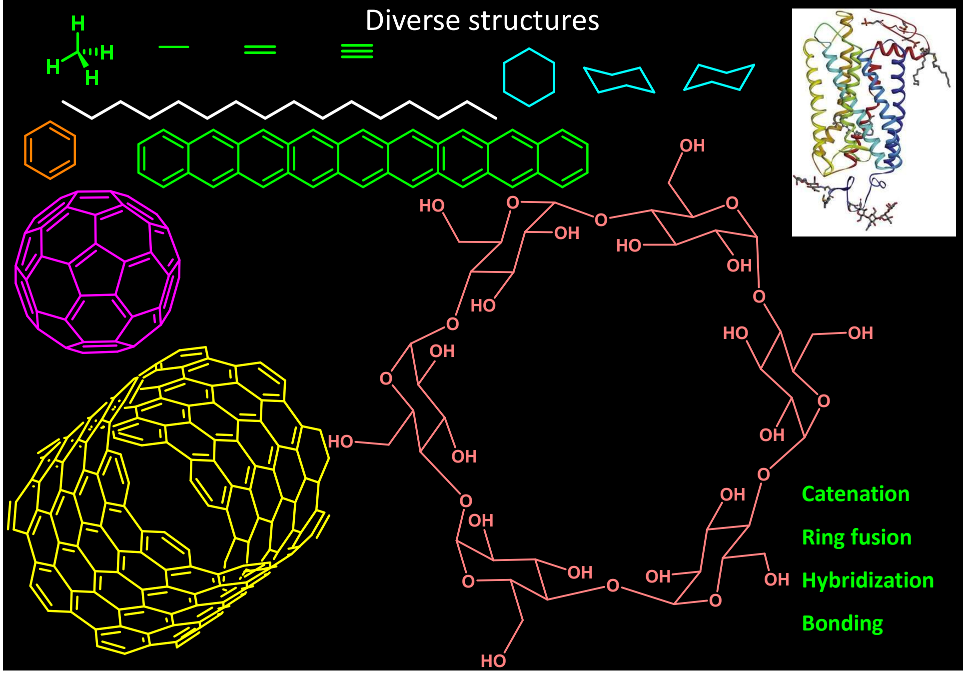


The Periodic Table According to Organic Chemists

1 H 1.0079																	2 C 12.011
3 C 12.011	4 C 12.011											5 C 12.011	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 C 12.011
11 C 12.011	12 C 12.011											13 C 12.011	14 C 12.011	15 C 12.011	S 32.065	17 Cl 35.453	18 C 12.011
19 C 12.011	20 C 12.011	21 C 12.011	22 C 12.011	23 C 12.011	24 C 12.011	25 C 12.011	26 C 12.011	27 C 12.011	28 C 12.011	29 C 12.011	30 C 12.011	31 C 12.011	32 C 12.011	33 C 12.011	34 C 12.011	35 Br 79.904	36 C 12.011
37 C 12.011	38 C 12.011	39 C 12.011	40 C 12.011	41 C 12.011	42 C 12.011	43 C 12.011	44 C 12.011	45 C 12.011	46 C 12.011	47 C 12.011	48 C 12.011	49 C 12.011	50 C 12.011	51 C 12.011	52 C 12.011	53 I 126.90	54 C 12.011
55 C 12.011	56 C 12.011	71 C 12.011	72 C 12.011	73 C 12.011	74 C 12.011	75 C 12.011	76 C 12.011	77 C 12.011	78 C 12.011	79 C 12.011	80 C 12.011	81 C 12.011	82 C 12.011	83 C 12.011	84 C 12.011	85 C 12.011	86 C 12.011
87 C 12.011	88 C 12.011	103 C 12.011	104 C 12.011	105 C 12.011	106 C 12.011	107 C 12.011	108 C 12.011	109 C 12.011	110 C 12.011								

Why organic chemistry?

Diverse structures



Catenation

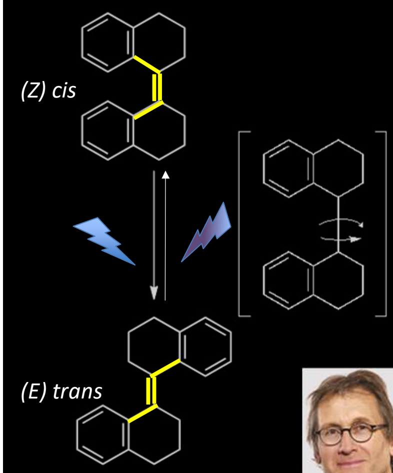
Ring fusion

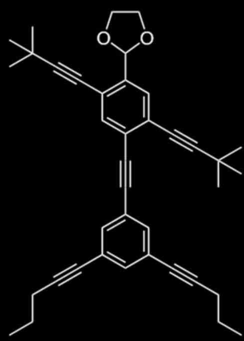
Hybridization

Bonding

Why organic chemistry?

Fantasy world





NanoPutian

2-(2,5-bis(3,3-dimethylbut-1-yn-1-yl)-4-{[3,5-di(pent-1-yn-1-yl)phenyl]ethynyl}phenyl)-1,3-dioxolane

Nomenclature

Isomerism

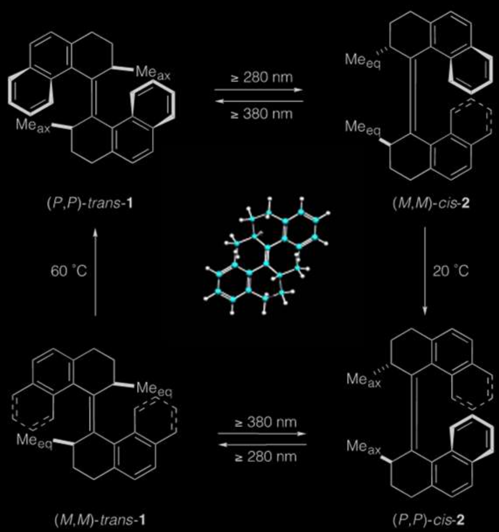
Photochemistry

Nature, **1999**, 401, 152

Why organic chemistry?

Fantasy world

Light-driven monodirectional molecular rotor



Nature, **1999**, 401, 152

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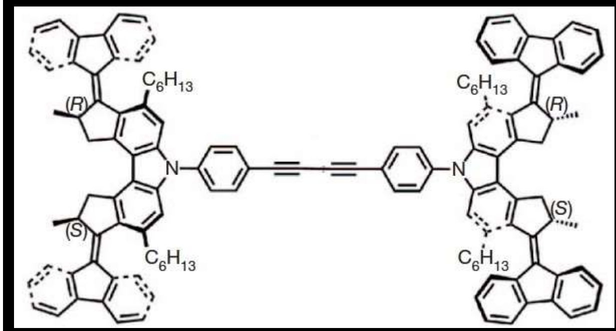
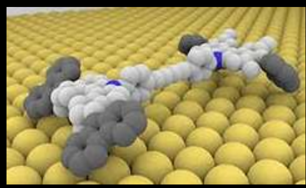
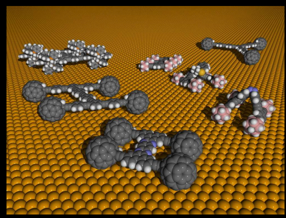
Why organic chemistry?

Fantasy world







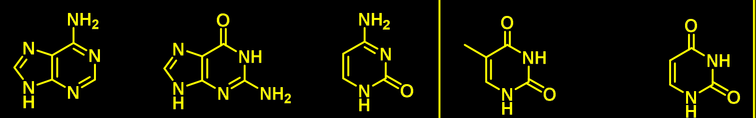
The "real" Nano Car

B. L. Feringa

Why organic chemistry?

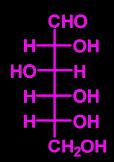
Nature's choice

Nucleobases

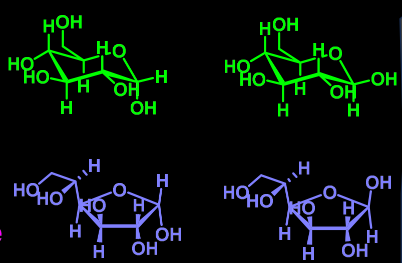


Adenine Guanine Cytosine Thymine Uracil

Carbohydrates

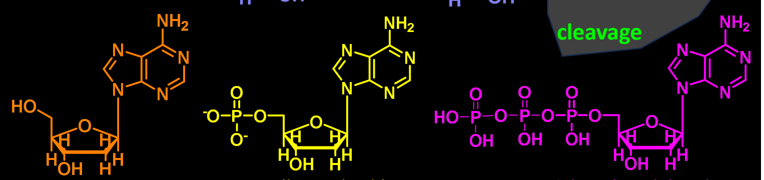


D-Glucose



- Heterocycles
- Conformations
- Ether linkage
- Energy storage and release by bond formation and cleavage

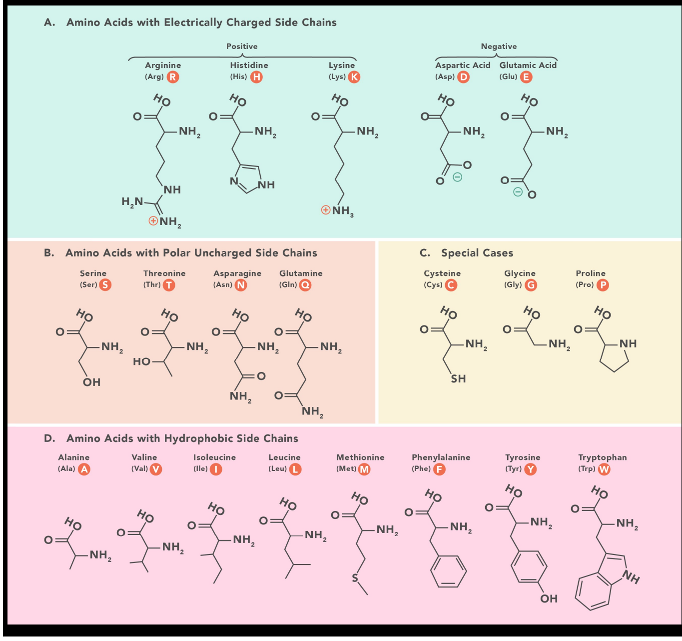
Nucleic acids
DNA and RNA



Deoxyribo-nucleoside Deoxyribo-nucleotide Adenosine triphosphate

Why organic chemistry?

Nature's choice

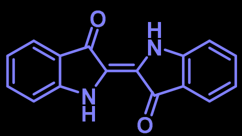


Amino acids
↓
Peptides
↓
Proteins
↓
Structure and functions

- Functional groups
- Amide bonds
- Hydrogen bonding
- Catalysis

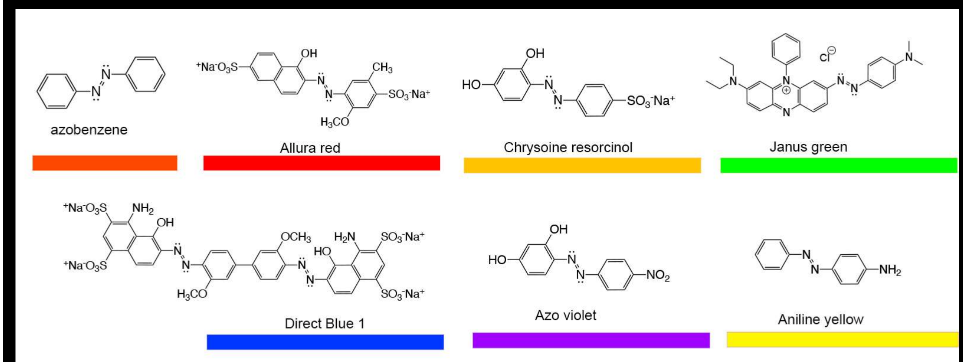
Why organic chemistry?

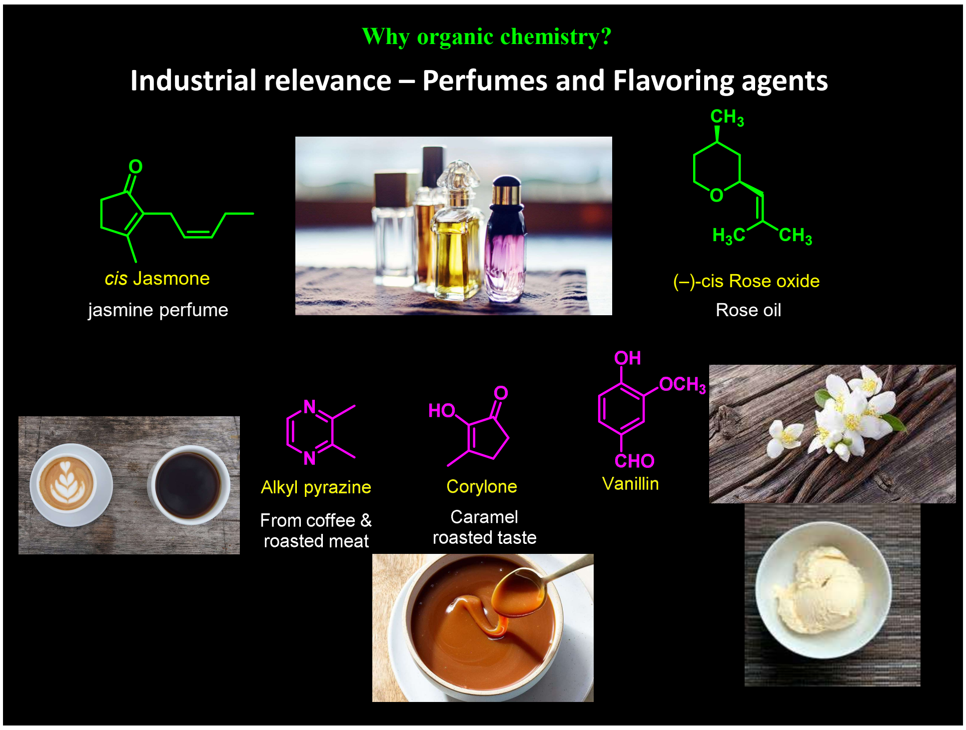
Industrial relevance – Dyes and pigments



Indigo

- Resonance
- Extended conjugation
- Light absorption
- Chromphores




Why organic chemistry?
Industrial relevance – Perfumes and Flavoring agents
O
cis Jasmone
jasmine perfume
CH3
O
H3C
CH3
(–)-cis Rose oxide
Rose oil
N
N
Alkyl pyrazine
From coffee &
roasted meat
HO
O
Corylone
Caramel
roasted taste
OH
OCH3
CHO
Vanillin